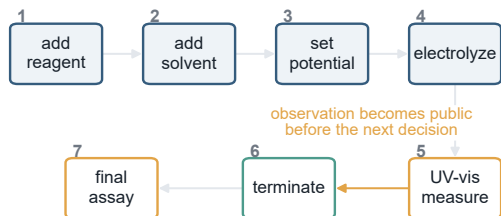


a One vessel exposes seven primitive operations



opaque condition · one simulated world · descriptive example

b Campaign receipts preserve units and denominators

closure 6 vessels · 6 closed · 6 final assays · 0 discards

actions 69 operation attempts · 17 non-final measurements

stocks 0.24 mol reagent · 0.48 L solvent

physical 2.763 cost · 39.0 h · 2.63 mL sample

trajectory-event alignment verified · resource receipt reconciled

c Identity, resources and replay stay external

■ **campaign identity**
world · condition
vessel · lifecycle

■ **resource authority**
budget contract
reconciled receipt

■ **event alignment**
trajectory ↔ receipt
verified

■ **exact replay**
60/60 lifecycles
10/10 paired cells

The ledger is not prompt context; it is the auditable measurement spine

d Closure and failure accounting are explicit

closed lifecycles **60 / 60**

final assays **60**

invalid or rejected **0**

right-censored **0**

815 accepted operations · 164 non-final measurements
repeated events, not independent sample units